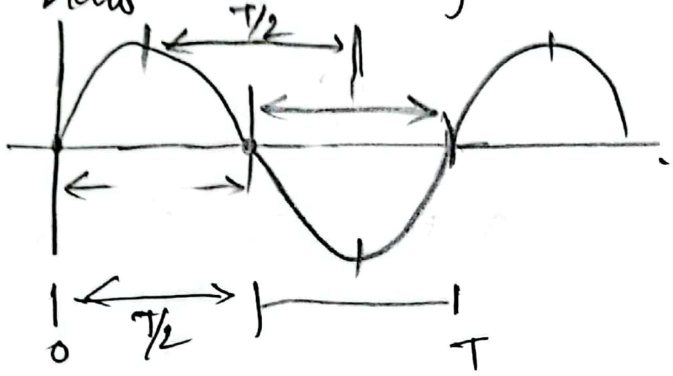
Assignment - 1

Md. Tashin Parvez.

ID: 011221437

Sec: C $\boxed{\text{Fall'23}}$

1) (i) Hence 2 point's have velocity 0 so we can say that the time given is for the half wave of a sine wave.



$\therefore T/2 = 0.25s.$

$\therefore T = 0.5s.$

(ii) freq. $F = 1/T$

$= 1/0.5$

$= 2Hz.$

(iii) Distance of 2 point is 36 cm.

So Amplitude $= 36/2$

$= 18 cm.$

2) (i) $\omega = \frac{2\pi}{T}$

$T = \frac{2\pi}{\omega}$ $\quad | \omega = \sqrt{k/m}$

$= 2\pi \sqrt{\frac{m}{k}}$

$= 2\pi \sqrt{\frac{0.68}{6.5}}$

$= 0.64 s.$

(ii) $\omega = \sqrt{k/m}$

$= \sqrt{65/0.68}$

$= 9.78$ rad/s.

(iii) phase constant $= 0°$
Because it started when it's in ~~its~~ its amplitude & $t=0$ position.

(iv) $x = A\cos(\omega t)$

$= 0.11\cos(9.78t)$ | $A = 11\,cm$ $= 0.11\,m$.

$V = dx/dt$

$= -0.11 \times 9.78\sin(9.78t)$

[4 sec] $V = -0.11 \times 9.78\sin(9.78 \times 4)$

$= -1.055\,ms^{-1}$

[max] $V = -A\omega$

$= -(0.11 \times 9.78)$.

$= -1.0758\,ms^{-1}$

(v) $a = dv/dt$

$= -0.11 \times 9.78 \times 9.78 \cos(9.78t)$

[4sec] $a = -0.11 \times (9.78)^2 \cdot \cos(9.78 \times 4)$

$= = -1.982\, ms^{-2}$

[max] $a = -10.52\, m/s^2$. $\left[a_{max} = -0.11 \times (9.78)^2 \right]$

(VI) [t=0] $x = 0.11 \cdot \cos(0)$

$= 0.11\, m.$ $[x = A]$

[t=7sec] $x = 0.11 \cdot \cos(9.78 \times 7)$

$= 0.087\, m.$

(VII)

$V_1 = \omega\sqrt{A^2 - x^2}$

$= 0\, ms^{-1}$

$A = 0.11\, m$

$x_1 = 0.11\, m.$

$\omega = 9.78.$

$V_2 = \omega\sqrt{A^2 - x^2}$

$= 1.002\, ms^{-1}$

$x_2 = 0.04\, m.$

(3)

(i) $a_{max} = A\omega^2$

$= A\left(2\pi/T\right)^2$

$= 83.89$

$F_{max} = ma$

$= 0.12 \times 83.89$

$= 10.06\ N$

(ii) $K = \omega^2 m$

$= m\left(2\pi/T\right)^2$

$= 118.435\ N/m.$

(4)

(i) $A = x/2 = \frac{2\times10^{-3}}{2}$

$= 0.001m.$

(ii) $V_{max} = A\omega.$

$= A \times 2\pi f$

$= 0.001 \times 2\pi \times 120\ Hz.$

$= 0.754\ ms^{-1}$

(iii) $a_{max} = A\omega^2$

$= A\left(2\pi f\right)^2$

$= 568.5\ ms^{-2}$

5)

a) $F = ma$

$-kx = ma$

$\left(\frac{K}{m}\right) = -a/x$

$\omega^2 = -\left(\frac{-123}{0.1}\right)$

$(2\pi f)^2 = 1230$

$f = \frac{\sqrt{1230}}{2\pi}$

$= 5.58\,Hz.$

(ii) from (1) $\omega^2 = 123/0.1$

$\omega = 35.07\ rad/s$

$\omega^2 = K/m$

$m = K/\omega^2$

$= \frac{400}{(35.07)^2}$

$= 0.325\,kg.$

(W)

(iii) $V = \omega\sqrt{A^2 - x^2}$

$A = \sqrt{\left(\frac{V}{\omega}\right)^2 + x^2}$

$= \sqrt{\left(\frac{-13.6}{35.07}\right)^2 + (0.1)^2}$

$= 0.4005\,m.$

8) (i) $\omega = \frac{2\pi}{T}$

$\sqrt{\frac{k}{m}} = 2\pi/T \Rightarrow T = 2\pi \times \sqrt{m/k}$

$= 1.57s.$

$m = 0.025\,kg$

(ii) $\omega = 2\pi f$

$\sqrt{k/m} = 2\pi f$

$f = \frac{1}{2\pi}\sqrt{k/m}$

$= 0.636\,Hz$

$K = 400$ dyns/cm.

$= 0.4\,N/m$

(iii) $\omega = \sqrt{k/m} = \sqrt{\frac{0.4}{0.025}}$

$= 4$ rad/s.

(iv) $V_{max} = -A\omega.$

$= \cancel{-(0.1 \times 0.25)}$

$= \cancel{-0.025}$ m/s.

$V_{max} = -A\omega$

$= -0.1 \times 4$

$= -0.4\,ms^{-1}$

(7) $F = ma$

$= m\omega^2 A$

$= m(2\pi f)^2 A$

$= 1.68\times10^{-27} \cdot (2\times\pi\times10^{14})^2 \cdot 10^{-10}$

$= 6.63\times10^{-8}\,N.$

(8) $V = V_{max} = A\omega$

$\therefore A\omega = 1\,m/s.$

$a = a_{max} = A\omega^2 = \boxed{A\cdot\omega}\,\omega$

$a = 1\cdot\omega.$

$\omega = a. = 1.57.$

$\therefore T = \frac{2\pi}{\omega}$

$= 4.0028.$

2)

(1) $F = Kx$

$ma = Kx$

$K/m = a/x$

$\omega^2 = a/x$

$\omega = \sqrt{a/x}$

$= \sqrt{48/3}$

$= 4\,rad/s.$

$V = \omega\sqrt{A^2 - x^2}$

$= 4\sqrt{5^2 - 3^2}$

$= 16\,ms^{-1}$

$$\omega = 2\pi/T$$

$$T = 2\pi/4$$

$$= 1.57 s.$$

(iii) $V_{max} = A\omega$

$$= 5 \times 4$$

$$= 20 ms^{-1}$$

10) (i) $T = 1/f$

$$= 4 s.$$

(ii) $\omega = \frac{2\pi}{T}$

$$= 1.57 \text{ rad/s}.$$

(iii) $A = x = 0.37 cm$ [$V = 0$ ∴ Amplitude is max

$$= 0.0037 m$$

(iv) $x_3 = A \cos \omega t$

$$= 0.0037 \times \cos(1.57 * 3)$$

$$= -8.8392 \times 10^{-6} m.$$

(A)

$V_3 = -A\omega \sin(\omega t)$

$= -(0.0037 \times 1.57) \sin(1.57 \times 3)$

$= -5.808 \times 10^{-3}\ ms^{-1}$

|||| (i) $x_{2.5} = 10 \cos(3\pi \times 2.5 + \pi/3)$

$= \cancel{0.5m}$

$= 8.66\ m.$

(ii) $V_{3.0} = -10 \times 3\pi \sin(3\pi t + \pi/3)$

$= 81.62\ m/s$

(iii) $a_2 = -10\ (3\pi)^2 \cos(3\pi t + \pi/3)$

$= -444.13\ ms^{-2}$

(iv) $V_{max} = -\omega A$

$= -94.24\ ms^{-1}$

| $\omega = 3\pi$
| $A = 10$

(12) (i) $\omega = 2\pi f$

$f = 10/2\pi$

$= 1.6\ Hz$

| $A = 10\ m.$
| $\omega = 10\ rad/s$

(ii) $T = 1/f$

$= 0.625 s.$

(iii) $X_{max} = A = 10 m.$

(iv) $V_{max} = A\omega$

$= 10 \times 10 = 100 ms^{-1}$

(v) $a_{max} = -A\omega^2$

$= -10 \times 10^2$

$= -1000 ms^{-1}$

13)

(i) $T = 1/f$

$= 2s.$

(ii) $\omega = 2\pi f$

$= \pi \ rad/s.$

$x = A\cos(\omega t + \phi)$

$= 4\cos(\pi t + \pi/4)$

(iii) $V_5 = -4\pi \sin(\pi t + \pi/4)$

$= 8.886 m/s^{-1}$

$a_5 = -4\pi^2 \cos(\pi t + \pi/4)$

$= 27.915 m/s^{-2}$

14

(i) $\omega = \sqrt{k/m}$

$= 9.9$ rad/s.

$T = 1/f = \frac{1}{1.57}$

$= 0.636$ s.

(ii) $f = \omega/2\pi = 1.57$ Hz.

(iii) $x = A\cos(\omega t)$

$= 0.05\cos(9.9t)$

$x_{max} = A = 5$ cm

$= 0.05$ m

15

(i) $F = Kx$

$mg = Kx$.

$K = \frac{mg}{x} = \frac{0.5 \times 9.8}{7 \times 10^{-2}}$

$= 70$ N m^{-1}

(ii). $\omega = \sqrt{k/m} = 11.83$ rad/s.

(iii) $\omega = 2\pi/T$.

$T = 2\pi/\omega = 0.531$ sec.

(iv) $E_p = \frac{1}{2}Kx^2$

$= \frac{1}{2} \times 70 \times (0.07)^2$

$= 0.1715$ J

$x_{max} = A$

(v) $E_K = \frac{1}{2}mv^2$

$= \frac{1}{2}(0.5)(0.4)^2$

$= 0.04$ J.

16/

$m = 25g = 0.025\,kg$.
$K = 400\,dynes/cm$
$= 0.4\,Nm^{-1}$
$x = 10\,cm = 0.1\,m$
$V = 40\,cms^{-1}$
$= 0.4\,m/s$.

(i) $T = 2\pi/\omega = 2\pi\sqrt{m/k}$
$= 1.57\,s$.

(ii) $f = 1/T = 0.637\,Hz$.

(iii) $\omega = \sqrt{k/m} = 4\,rad/s$.

(iv) $E_p = \frac{1}{2}kx^2$
$= 0.002\,J$

$E_K = \frac{1}{2}mv^2$
$= 0.002\,J$

$E_T = 4.0 \times 10^{-3}\,J$

(v) $E_T = \frac{1}{2}KA^2$
$A = \sqrt{2E/K} = 0.14\,m$.

(vi) $x = A\cos\theta_0$ $(\theta_0 = \omega t + \phi)$
$\theta_0 = 44.40^\circ$

(vii) $V_{max} = -A\omega$
$= -0.565\,ms^{-1}$

(viii) $a_{max} = -A\omega^2$
$= -2.256\,ms^{-2}$

(ix)

(ix) $\theta = \omega t + \phi$

$44.4 - \omega t = \phi$ [t=0]

$\phi = 44.4$

$\therefore x = 0.14 \cos(4t + 44.4)$

$v = -0.56 \sin(4t + 44.4)$

$a = -2.24 \cos(4t + 44.4)$.

(x) for $t = \pi/8$ s

$x = -0.098$ m

$v = -0.4$ m/s.

$a = 1.57$ ms^{-2}

(17) (i) $E_T = \frac{1}{2} K A^2$

$= \frac{1}{2} \omega^2 m A^2$

$= \frac{1}{2} 4\pi^2 f^2 m A^2$

$= 2\pi^2 \times 10^2 \times 2.72 \times 10^{-5} \times (0.2 \times 0.2)$

$= 2.15 \times 10^{-7}$ J

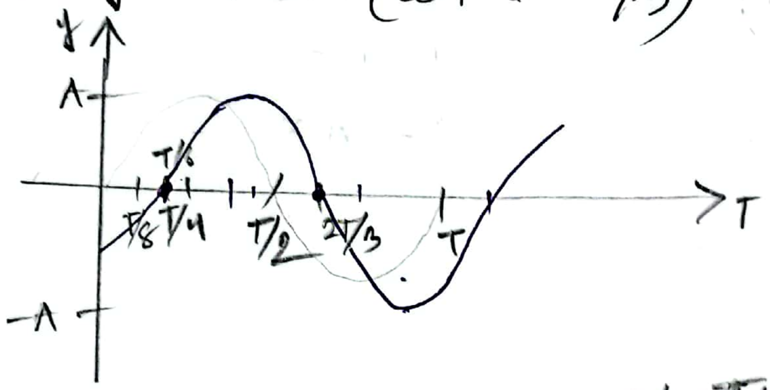
(ii) $V = V_{max}$ [t=0, Equlibrium pos]

$E = \frac{1}{2} m v^2$

$V_{max} = \sqrt{2E/m} = 12.57$ m/s.

(XV)

(18) (i) $y = A \sin(\omega t - \pi/3)$



$\omega t - \pi/3 = 0$

$\frac{2\pi}{T} t = \pi/3$

$T = \frac{2\pi t \times 3}{\pi}$

$= 6t$

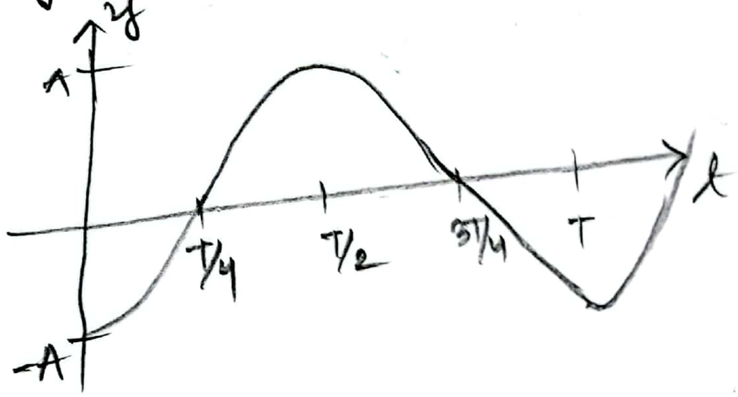
$t = T/6$

$T/6 + T/2$

$= \frac{T + 3T}{6} = 4T/6$

$= 2T/3$

(ii) $y = A \sin(\omega t - \pi/2)$



$\omega t - \pi/2 = 0$

$t = \pi/2 \times T/2\pi$

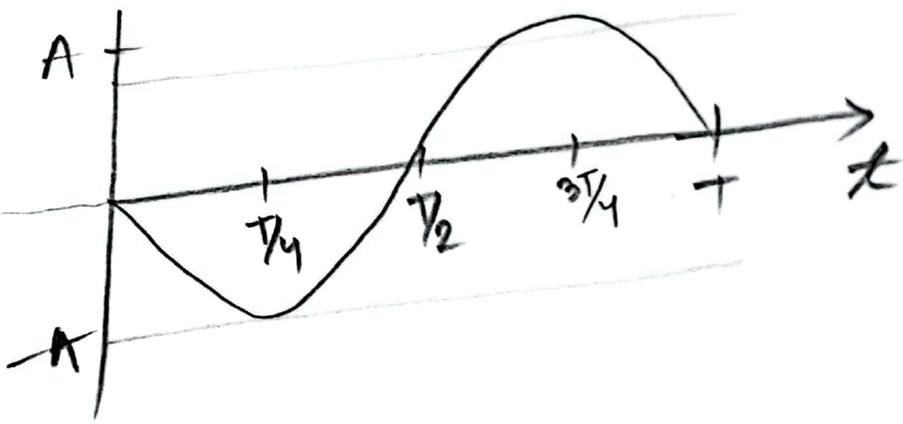
$= T/4$

(iii) $y = A \sin(\omega t + \pi)$

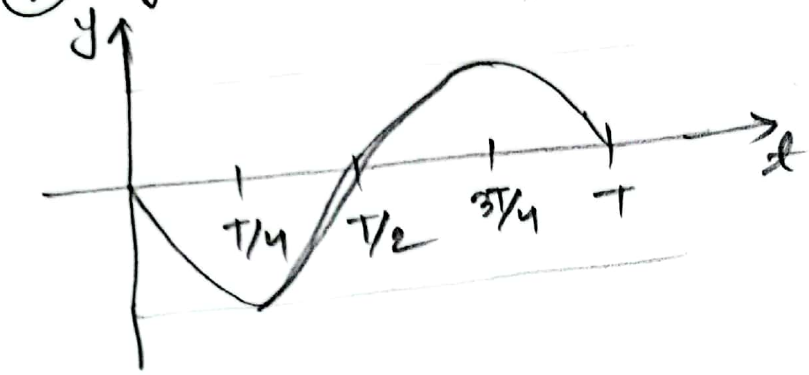
$\omega t + \pi = 0$

$t = -\pi \times \frac{T}{2\pi}$

$= -T/2$



(iv) $y = A\sin(\omega t - \pi)$



$$\omega t - \pi = 0$$
$$t = \pi \times \frac{T}{2\pi}$$
$$= T/2$$

(v) $y = A\sin(\omega t + 3\pi/4)$

$$\omega t + 3\pi/4 = 0$$
$$t = -\frac{3\pi}{4} \times \frac{T}{2\pi}$$
$$= -\frac{3T}{8}$$